

GRPO

Learn from a group of answers

Relative rewards without a separate critic



Replace a costly learned value baseline

PPO-style actor-critic

GRPO group baseline



Policy

+



Critic



→

r_1

r_2

r_3

Train an additional value model.

Use sampled rewards as the baseline.

More model / optimizer memory

No separate critic to train



Compare alternatives for the same prompt

Same question →

Reward 1

+

Reward 0

-

Reward 1

+

Reward 0

-

Above the group mean → positive advantage

Below the group mean → negative advantage

Compare within each prompt, rather than across unrelated tasks.



Improve the policy while limiting drift



Clipped update

+



Reference policy

Limit the policy-ratio incentive

KL penalty anchors behavior

Increase relative preference for better sampled responses

Retain reward scoring, clipping and reference regularization



Trade-off: multiple rollouts still cost compute.

Equal group rewards provide no relative reward signal.